DS 592: Introduction to Sequential Decision Making

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Homework 4
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Theory

Problem: Multi Dimensional Bandits

We define a multi-dimensional bandit problem where after pulling arm $i \in [K]$ the learner receives a d-dimensional random vector $\mathbf{r}_i \sim \mathbf{P}_i$. Each entry j of $\mathbf{r}_i$ which we denote as $\mathbf{r}_i(j)$ is β-subgaussian and $\mathbf{r}_i(j), \mathbf{r}_i(m)$ are independent from each other for $j \neq m$.

We will use the notation $\boldsymbol{\mu}_i$ to denote the d-dimensional vector of mean rewards of arm i so that $\boldsymbol{\mu}_i = \mathbb{E}_{\mathbf{r}_i \sim \mathbf{P}_i}[\mathbf{r}_i]$. We use the bold notation to denote vectors and other multi-dimensional objects.

Let $\mathbf{B} \subseteq \mathbb{R}^d$ be an arbitrary closed[1] set in R^d, and let $\boldsymbol{\mu}_\star = \min_{i \in [K]} \min_{\mu \in \mathbf{B}} \|\boldsymbol{\mu} - \boldsymbol{\mu}_i\|$, be the closest mean vector among all arms i to $\mathbf{B}$. We assume the following known bound on the norm of the differences,

$$\min_{\boldsymbol{\mu} \in \mathbf{B}} \|\boldsymbol{\mu} - \boldsymbol{\mu}_i\| \leq B$$

for some value $B > 0$ and for all $i \in [K]$.

We define the multi-dimensional regret with respect to $\mathbf{B}$ with norm $\|\cdot\|$ as

$$\mathrm{Regret}_{\mathbf{B}}^{\|\cdot\|}(n) = \mathbb{E}\left[\sum_{t=1}^{n} \min_{\boldsymbol{\mu} \in \mathbf{B}} \left\|\boldsymbol{\mu} - \boldsymbol{\mu}_{a_t}\right\| - \min_{\mu \in \mathbf{B}} \|\boldsymbol{\mu} - \boldsymbol{\mu}_\star\|\right].$$

Each term $\min_{\boldsymbol{\mu} \in \mathbf{B}} \left\|\boldsymbol{\mu} - \boldsymbol{\mu}_{a_t}\right\| - \min_{\mu \in \mathbf{B}} \|\boldsymbol{\mu} - \boldsymbol{\mu}_\star\|$ in the sum corresponds to the excess distance to $\mathbf{B}$ of action a_t w.r.t. to the optimal choice $\boldsymbol{\mu}_\star$.

Consider the case when the norm $\|\cdot\|$ equals $\|\cdot\|_\infty$ the ℓ_∞ norm defined as,

$$\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|$$

Design an algorithm that achieves a regret bound of order $\mathcal{O}\left(\beta\sqrt{Kn\log(Knd\max(B,1))}\right)$. Write your proofs in detail, specify what is the good event, and show it holds with high probability.

Hint: Design a UCB*-like algorithm where instead of computing an upper confidence bound of* $\min_{\boldsymbol{\mu} \in \mathbf{B}} \|\boldsymbol{\mu} - \boldsymbol{\mu}_i\|$ *you compute an* LCB *lower confidence bound.*

[1] This condition ensures the existence of $\arg\min_{\mu \in \mathbf{B}} \|\boldsymbol{\mu} - \boldsymbol{\mu}_i\|$ for all $i \in [K]$h.

Bonus Problems

Show the same algorithm (or a different one!) can also be used to obtain a bound for other choices of norms:

a) The ℓ_2 norm (Euclidean norm) for a vector $\mathbf{x} \in \mathbb{R}^n$ is defined as: $\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$

b) The ℓ_1 norm (Manhattan norm) for a vector $\mathbf{x} \in \mathbb{R}^n$ is defined as: $\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i| = |x_1| + |x_2| + \cdots + |x_n|$

Be careful to derive the sharpest possible dependence on d. Can you obtain better or worse exponents when considering different norms?

c) Derive an instance dependent analysis for the $\|\cdot\|_\infty$ algorithm you derived above.

Programming Assignment: Empirical Analysis of Multi-Armed Bandit Algorithms

Background

In this assignment, we will implement and evaluate two popular algorithms: Upper Confidence Bound (UCB) and Thompson Sampling (TS). You will plot the Bayesian Regret of Thompson Sampling under well specified and misspecified priors, and will also compare the regret of Thompson sampling and UCB in frequentist domains. You will see that TS can be used in frequentist scenarios even when there is no prior distribution over bandit tasks.

Part I - Bayesian Regret

Consider a K-Armed Gaussian Bandit problem such that,

$$\mathcal{P} = \mathcal{N}(\mathbf{0}_K, \mathbb{I}_K)$$

where $\mathbf{0}_K$ is the K-dimensional zero-vector and $\mathbb{I}_K$ is the K-dimensional identity matrix. The mean reward vector $\boldsymbol{\mu} \sim \mathcal{P}$. Once defined, the reward distribution of arm i is a gaussian distribution with variance 1 and mean value $\boldsymbol{\mu}(i)$ so that $P_i = \mathcal{N}(\boldsymbol{\mu}(i), 1)$.

Tasks

Consider a Bayesian Bandit scenario with $K = 10$ and task generation distribution defined above. For different values of the time horizon n you will plot the Bayes Regret of TS by running 50 independent simulations. At the start of each simulation you must sample an

"environment" mean reward vector from the prior $\mathcal{P}$, and then run TS under two different assumptions,

1. The TS algorithm knows the true prior $\mathcal{P}$. This is known as a well-specified setting.
2. The TS algorithm is computing its posterior sampling rule under the *wrong* assumption that $\mathcal{P}' = \text{Uniform}([-1, 1]^K)$ is the correct prior. Assume the final reward distributions are still Gaussian with unit variances. This is known in the literature as *misspecification*.

Run simulations for different time horizons $n = 100, 1000, 10000$ and present them in a single plot with two curves, one tracking the Bayes Regret of TS using the true prior $\mathcal{P}$ and the other where TS is using the misspecified prior $\mathcal{P}'$. In your plot, the x axis must correspond to values of n, and the y axis to the Bayesian Regret.

What do you observe? Is the Bayes Regret under a misspecified prior competitive with the Bayes Regret under the correct prior assumption?

In your final answer, write down the posterior sampling rule that you derived when using these different prior assumptions.

Part II - "Frequentist" Regret

Consider a ten-armed Gaussian bandit problem with the following parameters:

- Arm 1 (optimal): $\mu_1 = 1/2$, $\sigma^2 = 1$
- Arm 2 (second-optimal): $\mu_2 = 1/2 - \Delta$, $\sigma^2 = 1$
- Arms 3-10: $\mu_2 = -1/2$, $\sigma^2 = 1$
- Time horizon: $n = 2000$
- Gap parameter: $\Delta \in \{0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$

Tasks;

Implementation

1. Implement the following bandit algorithms:
   - UCB with the confidence interval choices that we learned in class.
   - TS with the gaussian prior assumption from Part I, i.e. means zero and unit variances..
2. For each algorithm, implement a function that:

- Takes parameters Δ, n, and number of simulations
- Returns the empirical average regret and its standard error

Monte Carlo Simulations

1. For each algorithm and each value of Δ:
   - Run 10 independent simulations
   - Compute the average regret
   - Compute the standard error of your estimate
2. Create a plot showing:
   - x-axis: values of Δ
   - y-axis: expected regret
   - One curve for each algorithm
   - Error bars (they might be too small to be visible)

Compare the empirical performance of UCB and Thompson Sampling

- Which algorithm performs better in which scenarios?
- How does the performance gap change over time?

Implementation Notes

- Use NumPy's random number generator with a fixed seed for reproducibility
- Initialize your algorithms properly:
  - For UCB, ensure all arms are pulled at least once
  - For Thompson Sampling, set appropriate prior distributions
- Compute regret as: $r_t = \mu^* - \mu_{a_t}$ where μ^* is the mean of the optimal arm and μ_{a_t} is the mean of the arm selected at time t

Submission Requirements

- Well-documented Python code (preferably in Jupyter notebook format)
- Clear implementation of both algorithms
- Plots in publication quality (with legends, labels, error bands)
- Analysis text discussing your findings
- README file explaining how to run your code

Grading Criteria

- Correctness of implementation (40%)
- Quality of experimental setup and analysis (30%)
- Text answer clarity and insights (30%)